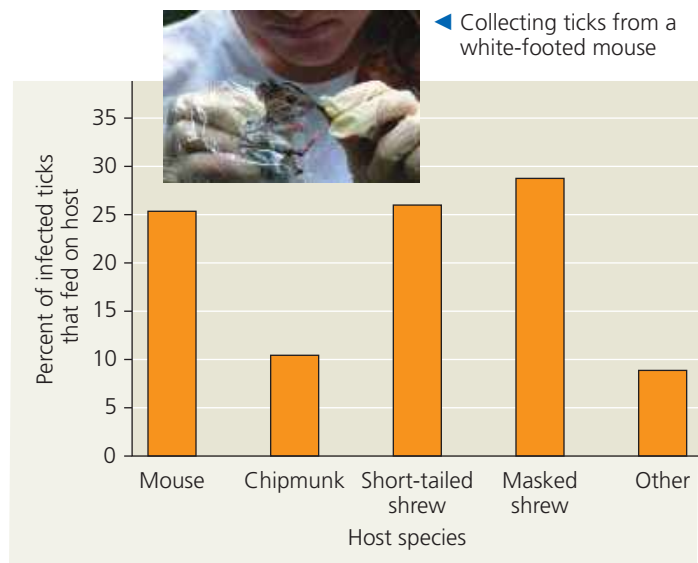


▼ **Figure 54.34 Unexpected hosts of the Lyme disease pathogen.** A combination of ecological data and genetic analyses enabled scientists to show that more than half of ticks carrying the Lyme pathogen became infected by feeding on the northern short-tailed shrew (*Blarina brevicauda*) or the masked shrew (*Sorex cinereus*).



MAKE CONNECTIONS Concept 23.1 discusses the significance of genetic variation between populations. How might genetic variation between shrew populations in different locations affect the number of infected ticks?

► **Figure 54.35 Tracking avian flu.** A graduate student bands a young gyrfalcon as part of a project to monitor the spread of avian flu.



the virus by trapping and testing migrating and resident birds in Alaska (Figure 54.35).

While our emphasis here has been on community ecology, pathogens are also greatly influenced by changes in the physical environment. To control pathogens and the diseases they cause, scientists need an ecosystem perspective—an intimate knowledge of how the pathogens interact with other species and with all aspects of their environment. Ecosystems are the subject of Chapter 55.

CONCEPT CHECK 54.5

1. What are pathogens?
2. **WHAT IF?** Rabies, a viral disease in mammals, is not currently found in the British Isles. If you were in charge of disease control there, what practical approaches might you employ to keep the rabies virus from reaching these islands?

For suggested answers, see Appendix A.

54 Chapter Review



➔ Go to **Mastering Biology** for Assignments, the eText, the Study Area, and Dynamic Study Modules.

SUMMARY OF KEY CONCEPTS

➔ To review key terms, go to the **Vocabulary Self-Quiz** in the **Mastering Biology** eText or Study Area, or go to goo.gl/zkjz9t.

CONCEPT 54.1

Interactions between species can help, harm, or have no effect on the individuals involved (pp. 1215–1221)

- **Interspecific interactions** affect the survival and reproduction of the individuals that engage in them. As shown in the table, these interactions can be grouped into three broad categories: competition, exploitation, and positive interactions.
- **Competitive exclusion** states that two species whose members compete for the same resource cannot coexist permanently in the same place. **Resource partitioning** is the differentiation of **ecological niches** that enables species to coexist in a **community**.

? For each interaction listed in the table, give an example of a pair of species that exhibit the interaction.

Interaction	Description
Competition (–/–)	Individuals of different species each use a limited resource, reducing the survival or reproduction of both individuals.
Exploitation (+/–)	Members of one species benefit by feeding upon (and thereby harming) members of the other species. Exploitation includes the following:
Predation	An individual of one species, the predator, kills and eats an individual of the other, the prey.
Herbivory	An herbivore eats part of a plant or alga.
Parasitism	The parasite derives its nourishment from a second organism, its host.
Positive interactions (+/+ or 0/+)	Members of one species benefit, while members of the other benefit or are not harmed. Positive interactions include the following:
Mutualism (+/+)	Members of both species benefit from the interaction.
Commensalism (+/0)	Members of one species benefit, while members of the other are not affected.